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 Studierichting: Informatica
 Oefeningenreeks 2D nr. 17.2

$$\begin{aligned}
 f(x) &= \frac{1 + \cos 2x}{\sin x} \\
 &= \frac{1 + (1 - 2 \sin^2 x)}{\sin x} \\
 &= \frac{2 - 2 \sin^2 x}{\sin x} \\
 &= 2 \frac{1 - \sin^2 x}{\sin x} \\
 &= 2 \frac{\cos^2 x}{\sin x}
 \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow \frac{\pi}{2}} f(x) &= 2 \frac{\cos^2\left(\frac{\pi}{2}\right)}{\sin\left(\frac{\pi}{2}\right)} \\
 &= 2 \frac{0}{1} \\
 &= 0
 \end{aligned}$$

$$\text{Voor } x \rightarrow 0 : \lim_{x \rightarrow 0} \left(\frac{1 + \cos(2x)}{\sin(x)} \right) = \frac{1 + \cos(0)}{\sin(0)} = \frac{1 + 1}{0} = \infty$$

$$\lim_{x \rightarrow 0+} \left(\frac{1 + \cos(2x)}{\sin(x)} \right) = \frac{1 + \cos(0+)}{\sin(0+)} = \frac{1 + 1+}{0+} = +\infty$$

$$\lim_{x \rightarrow 0-} \left(\frac{1 + \cos(2x)}{\sin(x)} \right) = \frac{1 + \cos(0-)}{\sin(0-)} = \frac{1 + 1+}{0-} = -\infty$$

$$\text{Voor } x \rightarrow \pi : \lim_{x \rightarrow \pi} \left(\frac{1 + \cos(2x)}{\sin(x)} \right) = \frac{1 + \cos(2\pi)}{\sin(\pi)} = \frac{1 + 1}{0} = \infty$$

$$\lim_{x \rightarrow \pi+} \left(\frac{1 + \cos(2x)}{\sin(x)} \right) = \frac{1 + \cos(2\pi+)}{\sin(\pi+)} = \frac{1 + 1+}{0-} = -\infty$$

$$\lim_{x \rightarrow \pi-} \left(\frac{1 + \cos(2x)}{\sin(x)} \right) = \frac{1 + \cos(2\pi-)}{\sin(\pi-)} = \frac{1 + 1+}{0+} = +\infty$$

References